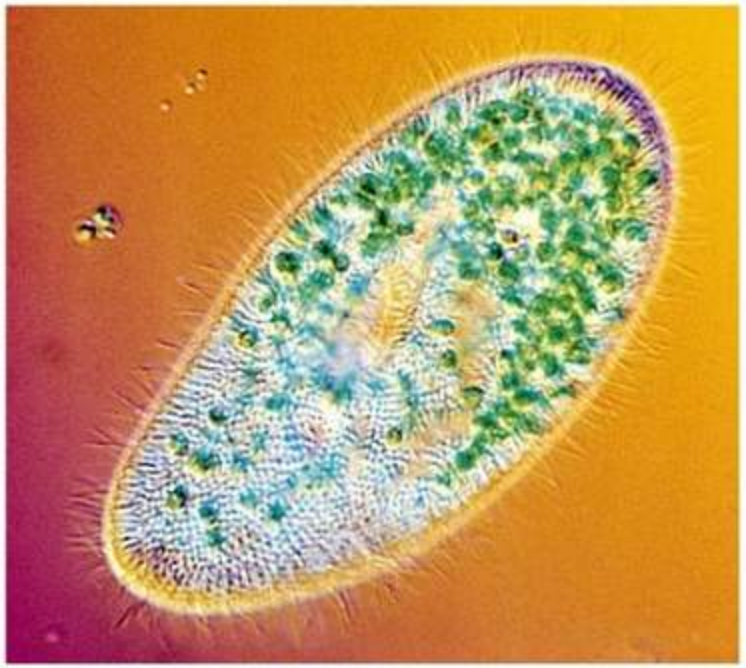
Study of Cell in relation to Biochemistry

Professor Rukhsana Rubeen,

Cells

- Smallest living unit
- Most are microscopic



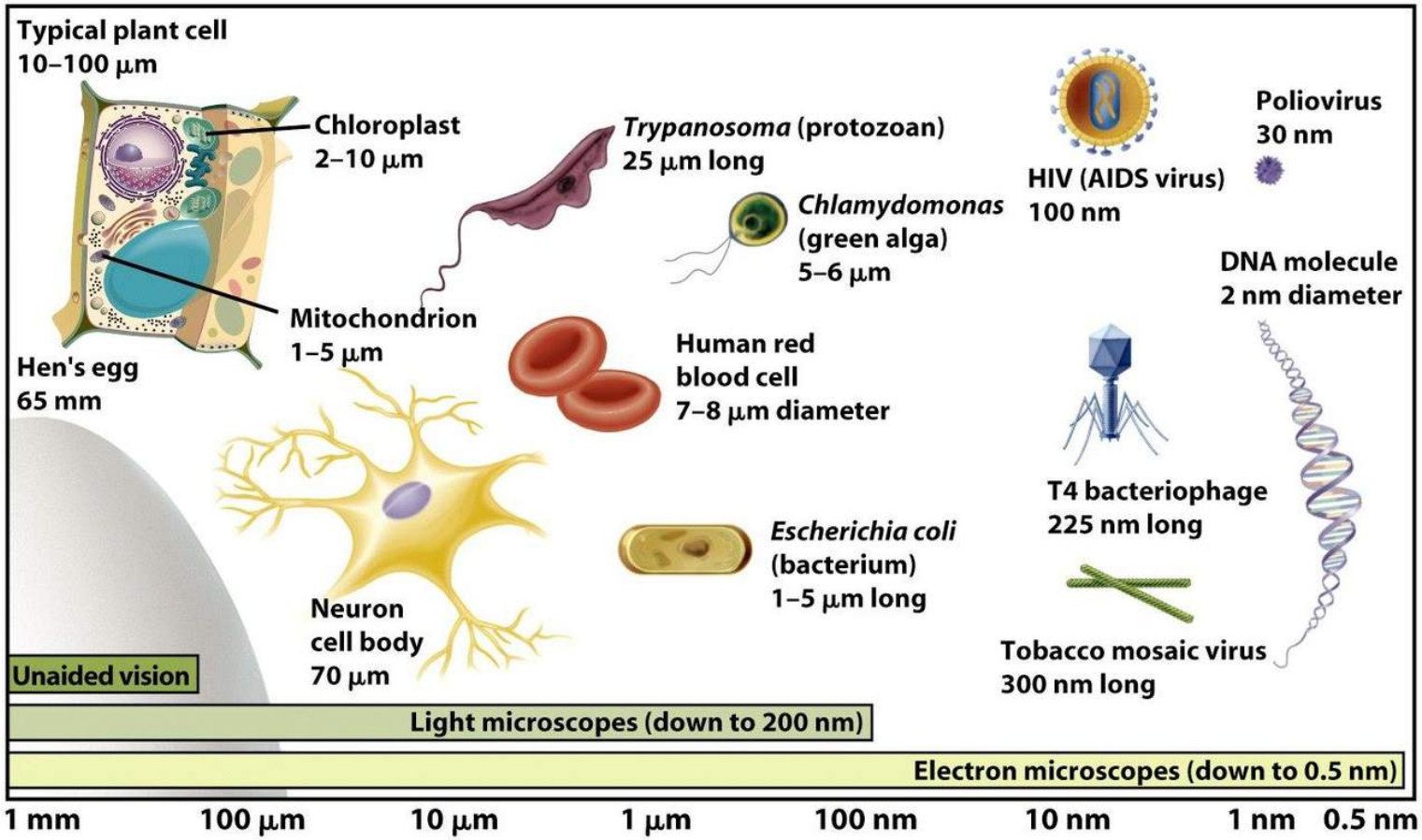


Principles of Cell Theory

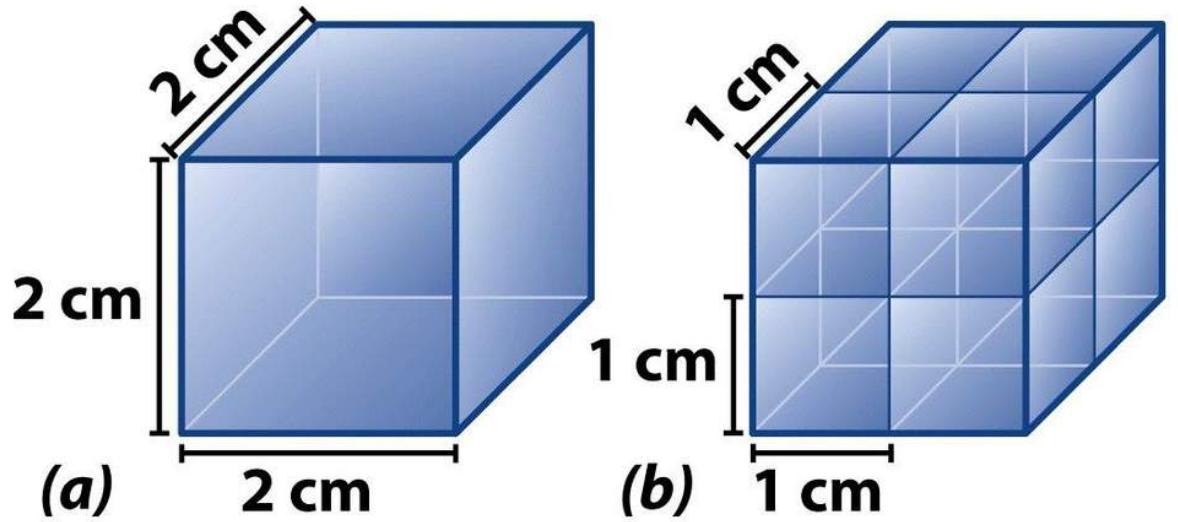
- All living things are made of cells
- Smallest living unit of structure and function of all organisms is the cell
- All cells arise from preexisting cells

(this principle discarded the idea of spontaneous generation)

Cell Size


Typical plant cell
10–100 μm
Chloroplast
2–10 μm
Mitochondrion
1–5 μm
Hen's egg
65 mm
Trypanosoma (protozoan)
25 μm long
Chlamydomonas
(green alga)
5–6 μm
Human red
blood cell
7–8 μm diameter
Neuron
cell body
70 μm
Escherichia coli
(bacterium)
1–5 μm long
HIV (AIDS virus)
100 nm
Poliovirus
30 nm
DNA molecule
2 nm diameter
T4 bacteriophage
225 nm long
Tobacco mosaic virus
300 nm long
Unaided vision
Light microscopes (down to 200 nm)
Electron microscopes (down to 0.5 nm)
1 mm
100 μm
10 μm
1 μm
100 nm
10 nm
1 nm
0.5 nm

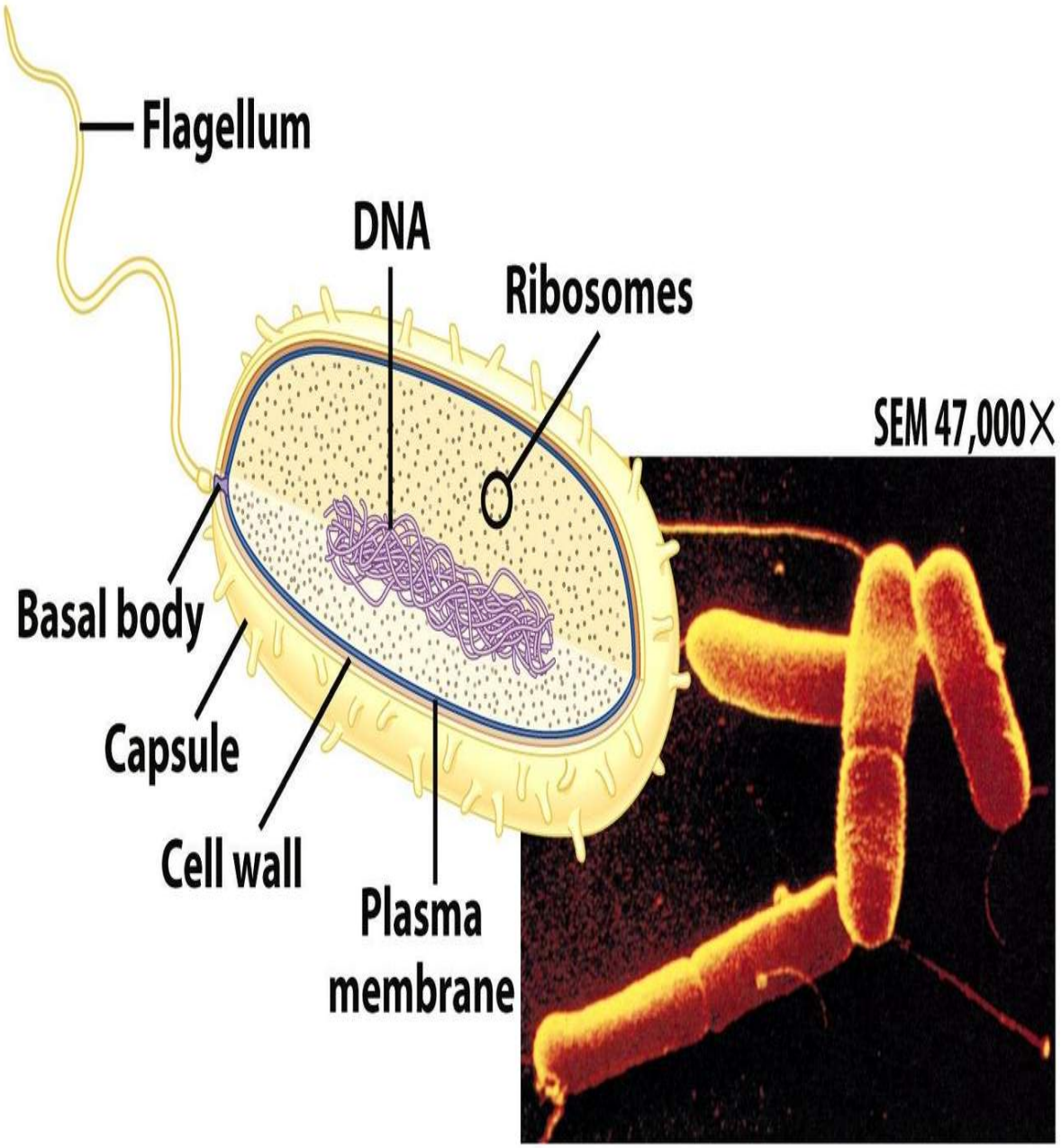
Cells Have Large Surface Area-to-Volume Ratio



Number of cells	1	8
Total surface area	24 cm^2	48 cm^2
Total volume	8 cm^3	8 cm^3
Surface area/volume	24/8 = 3:1	48/8 = 6:1

Prokaryotic Cells

- First cell type on earth
- Cell type of Bacteria and Archaea
- No membrane bound nucleus
- Nucleoid = region of DNA concentration
- Organelles not bound by membranes

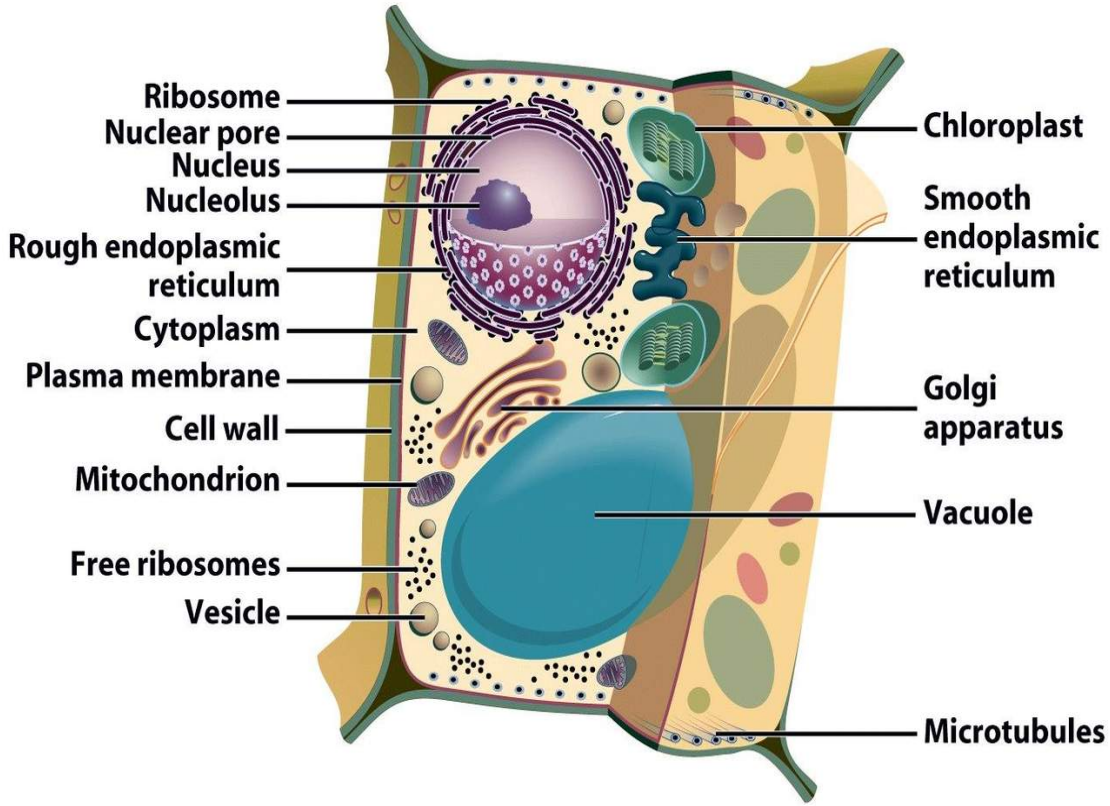


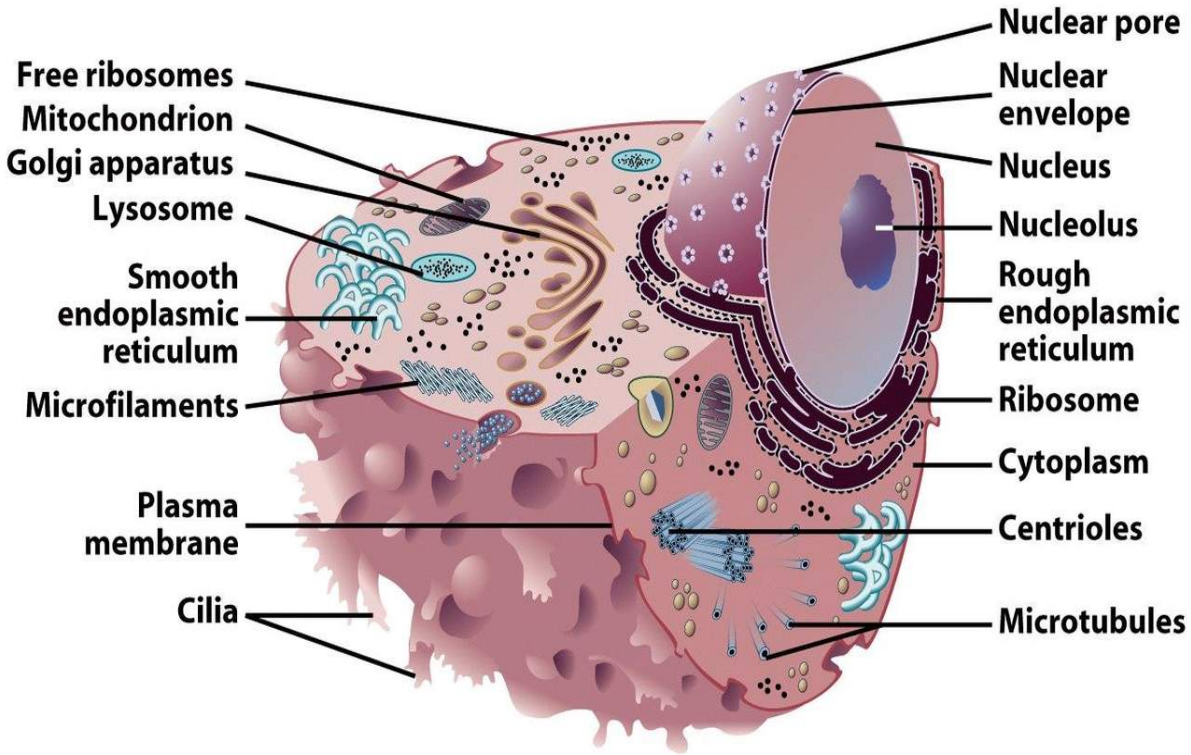
Eukaryotic Cells:

Nucleus bound by membrane

Include fungi, protists, plant, and animal cells

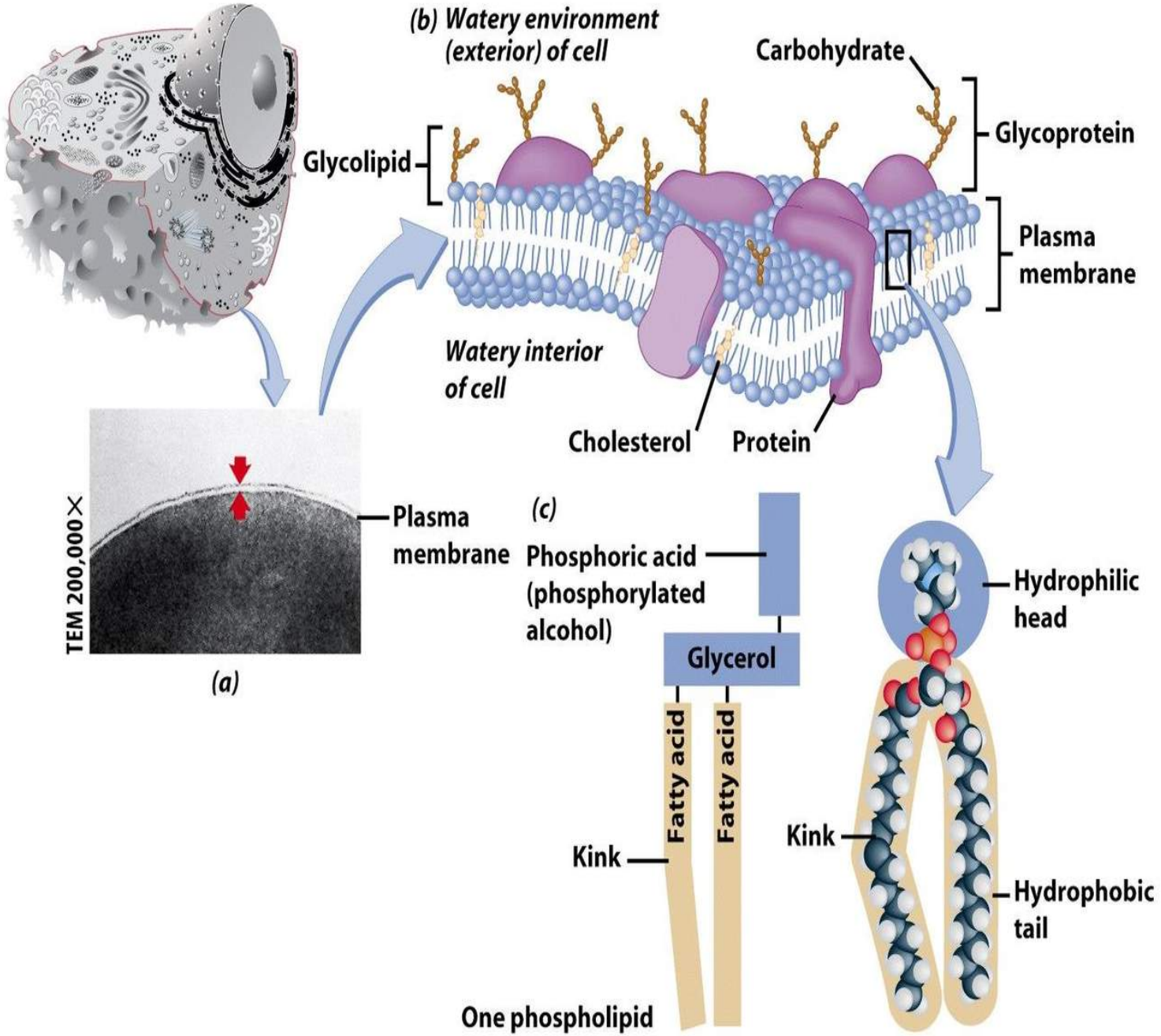
Possess many organelles



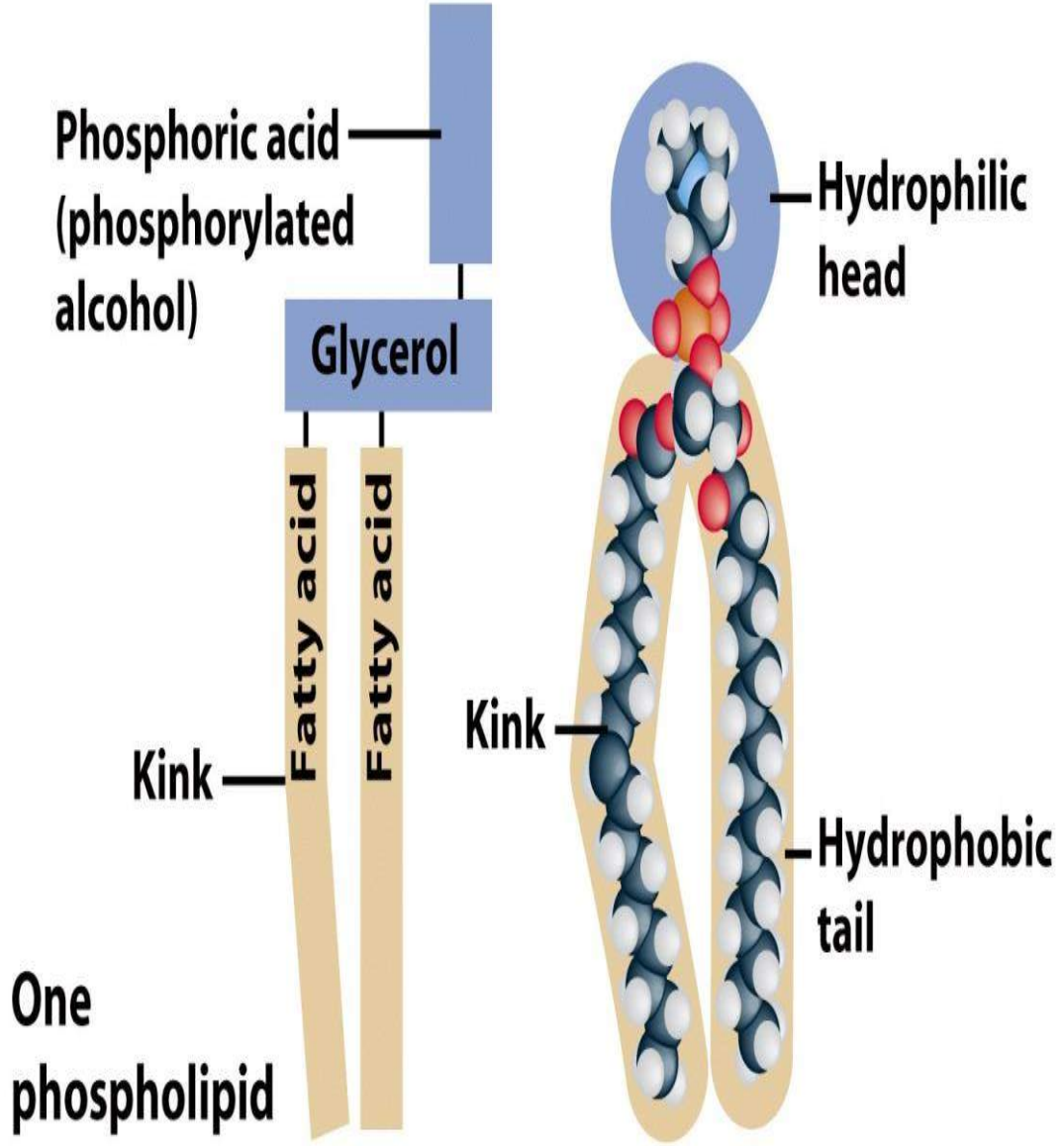


- **Plasma Membrane**

- Contains cell contents
- Double layer of phospholipids & proteins

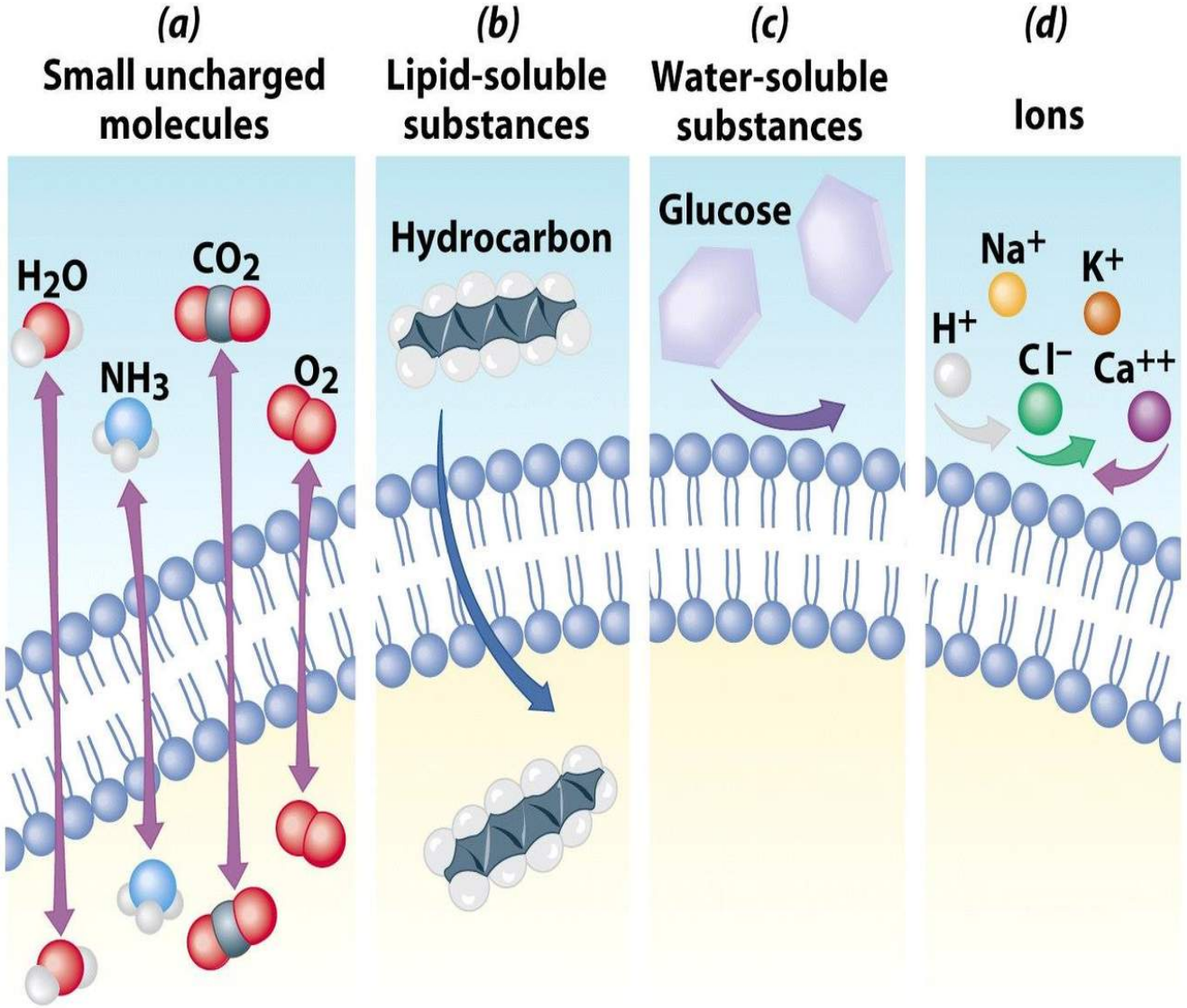


- **Phospholipids**
- Polar
  - Hydrophylic head
  - Hydrophobic tail
- Interacts with water



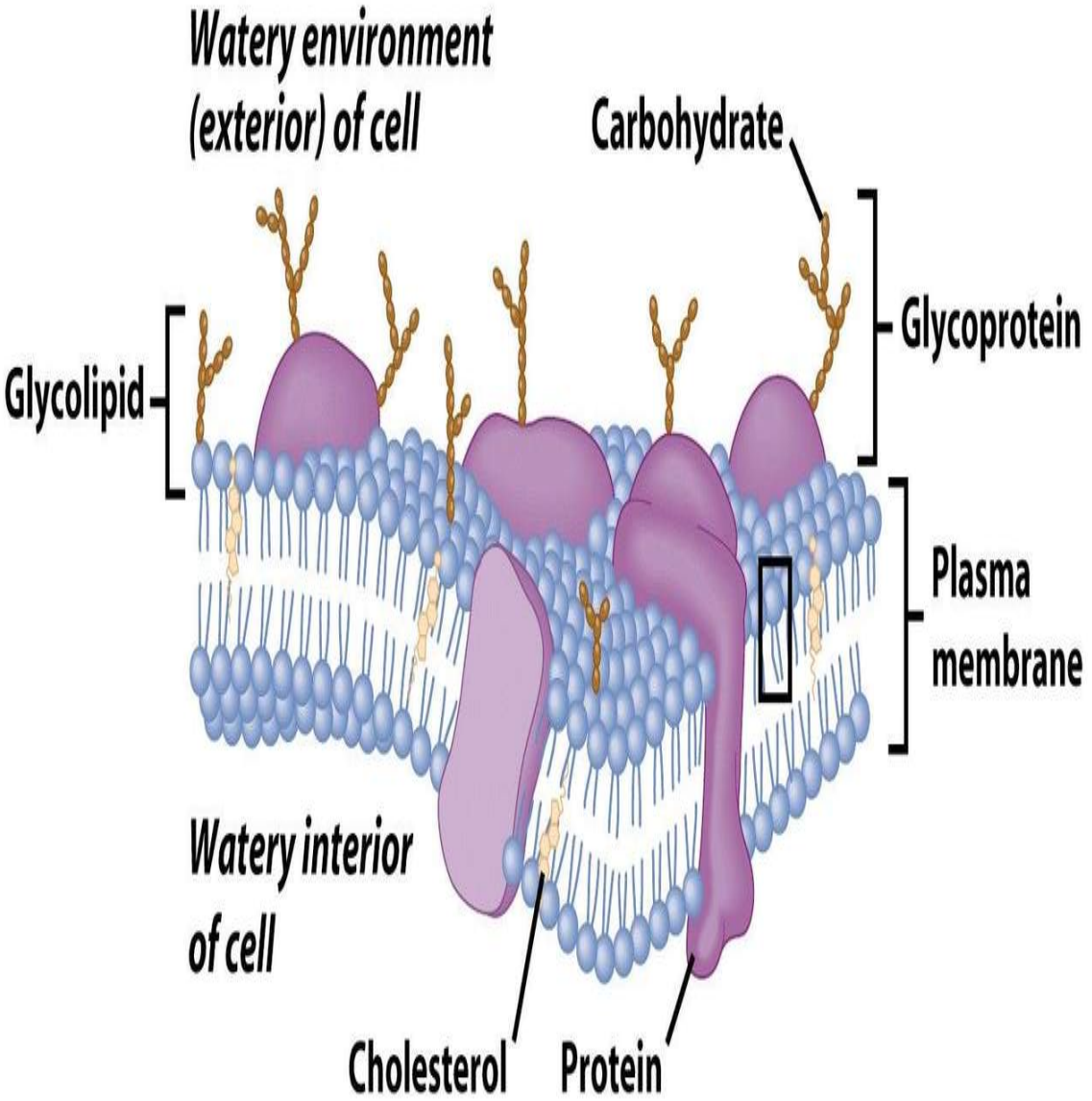
Movement Across the Plasma Membrane

- A few molecules move freely
  - Water, Carbon dioxide, Ammonia, Oxygen
- Carrier proteins transport some molecules
  - Proteins embedded in lipid bilayer
  - Fluid mosaic model – describes fluid nature of a lipid bilayer with proteins



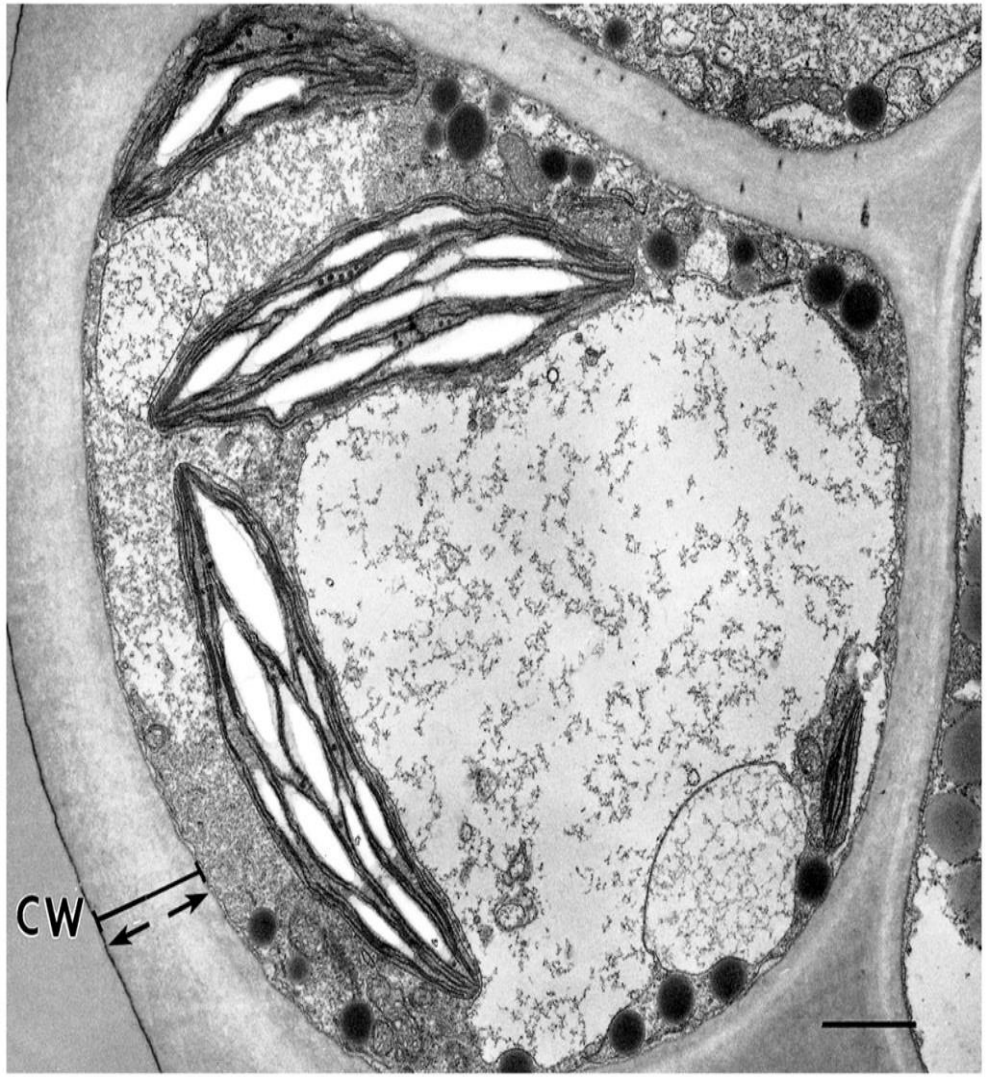
Membrane Proteins

- 1. Channels or transporters
  - Move molecules in one direction
- 2. Receptors
  - Recognize certain chemicals
- 3. Glycoproteins
  - Identify cell type
- 4. Enzymes
  - Catalyze production of substances



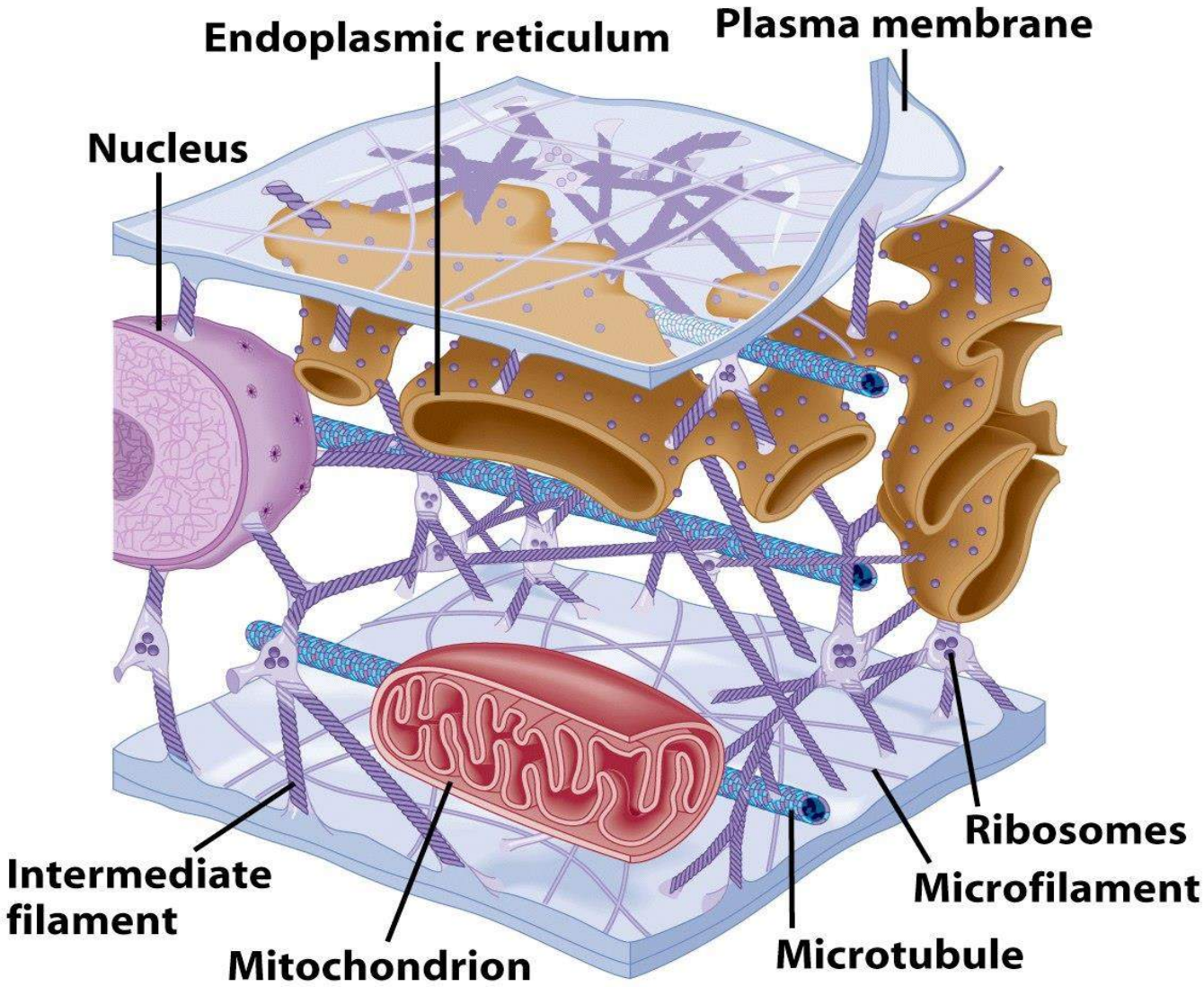
Cell Wall & Differences

- Found in plants, fungi, & many protists
- Surrounds plasma membrane
- Plants – mostly cellulose
- Fungi – contain chitin



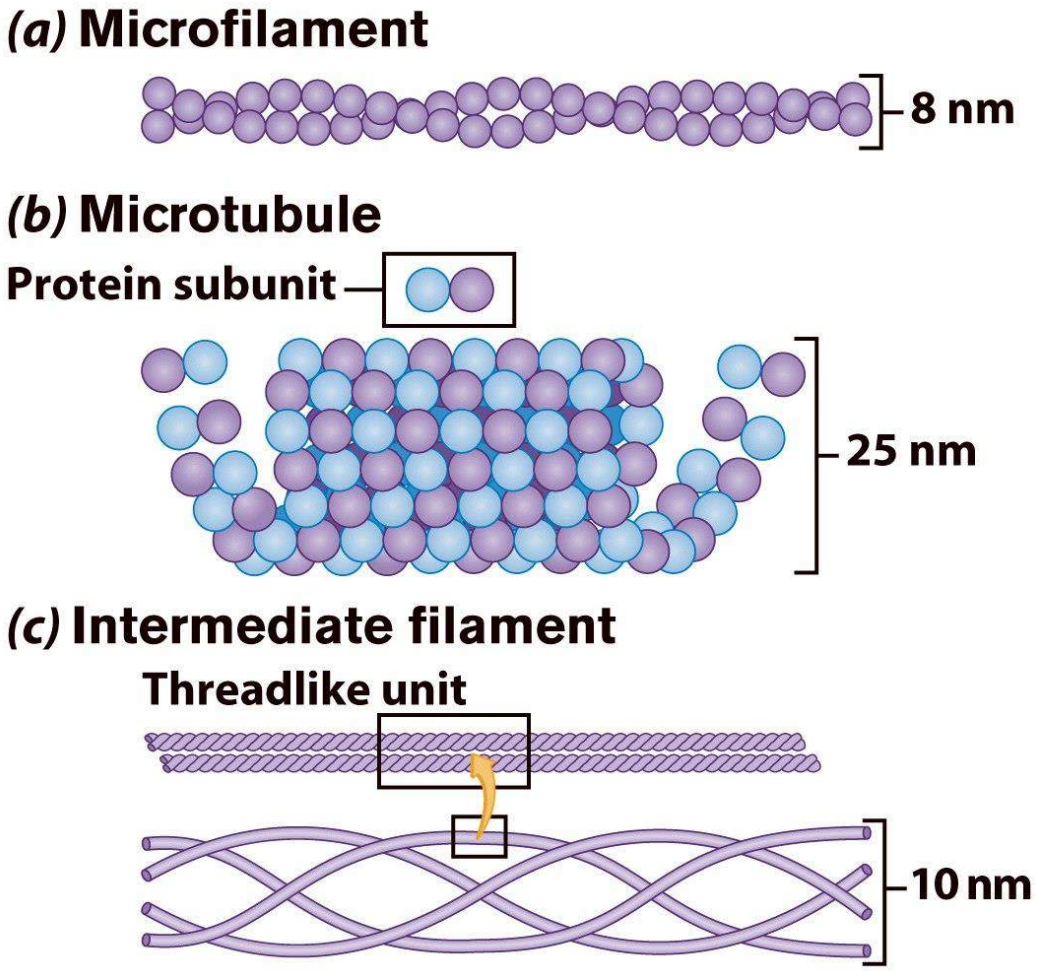
Cytoplasm

- Viscous fluid containing organelles
- components of cytoplasm
  - Interconnected filaments & fibers
  - Fluid = cytosol
  - Organelles (not nucleus)
  - storage substances



Cytoskeleton

- Filaments & fibers
- Made of 3 fiber types
  - Microfilaments
  - Microtubules
  - Intermediate filaments
- 3 functions:
  - mechanical support
  - anchor organelles
  - help move substances



Nuclear Envelope

- Separates nucleus from rest of cell Double membrane

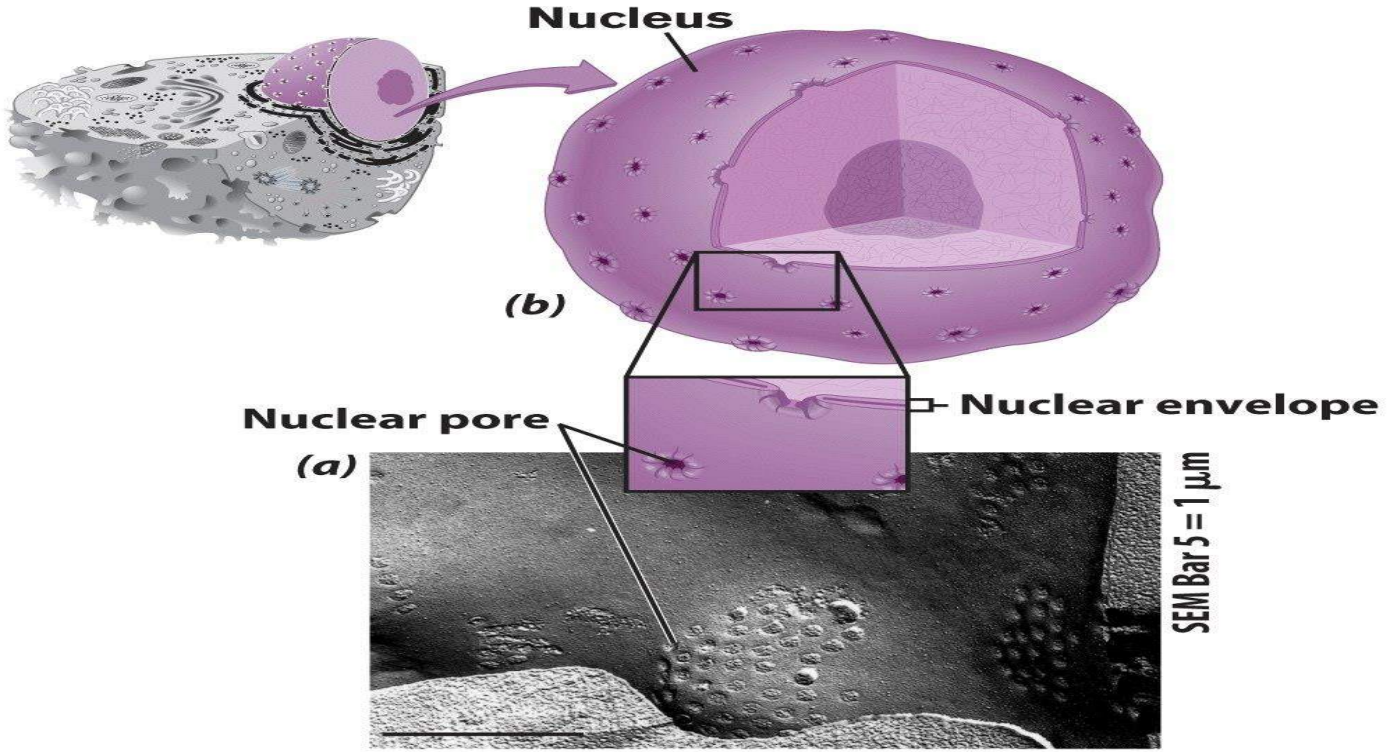
Pores

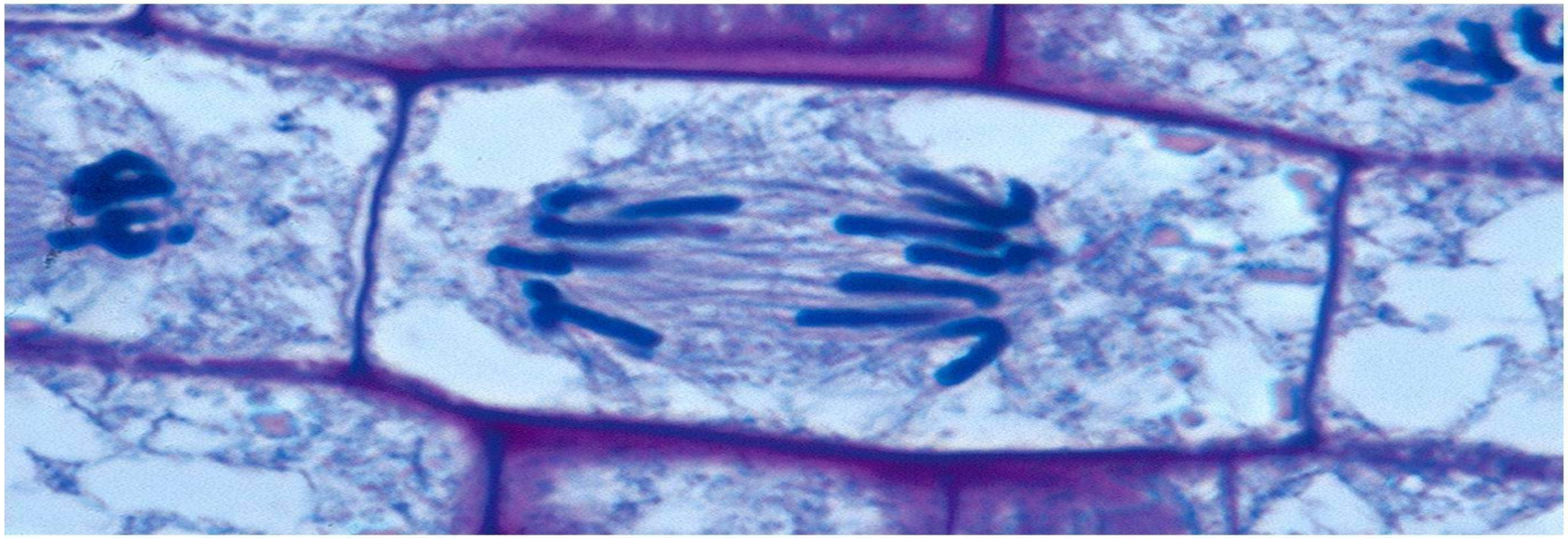
- DNA: Hereditary material

Chromosomes

  - DNA
  - Protiens
  - Form for cell division

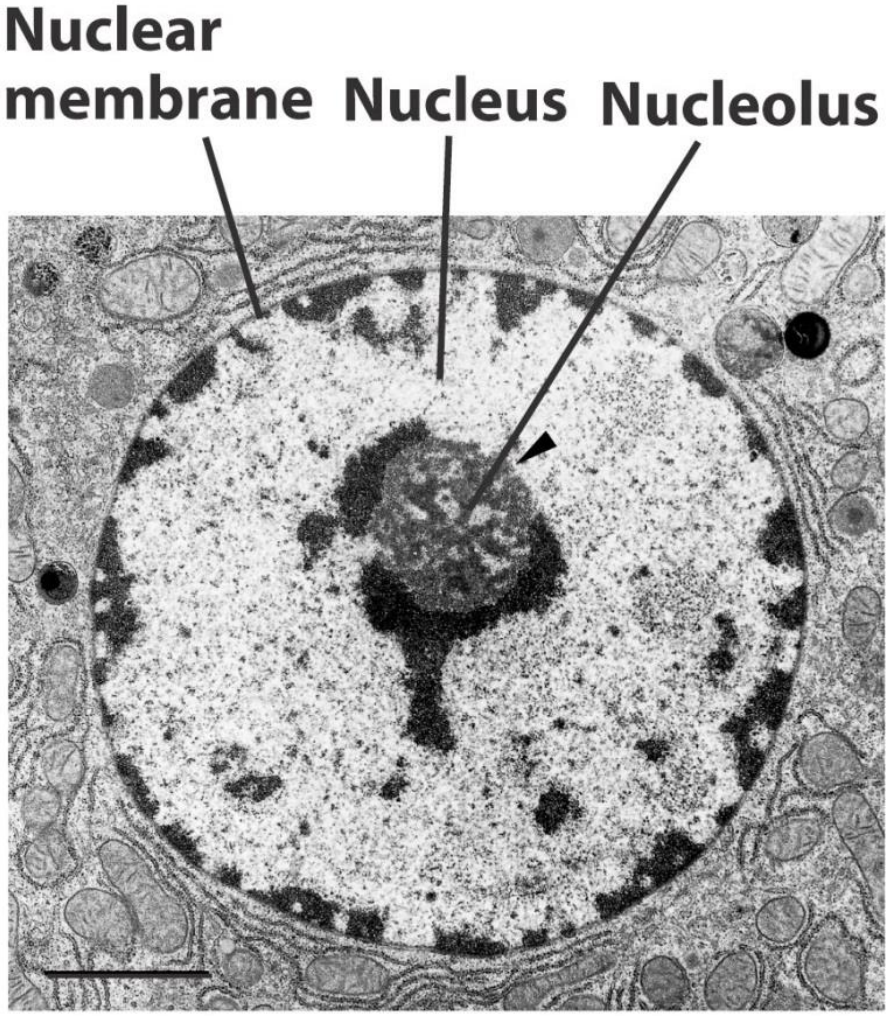
- Chromatin





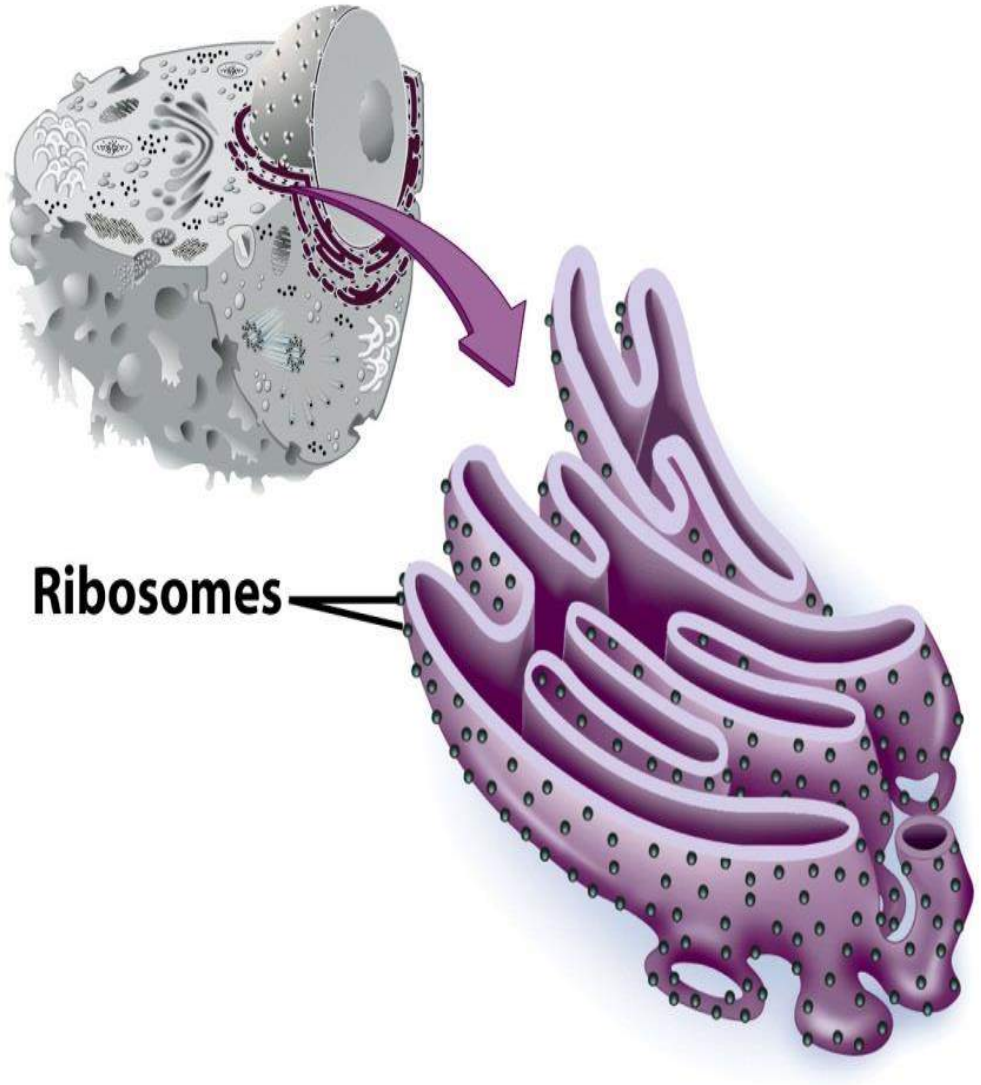
Nucleolus

- Most cells have 2 or more
- Directs synthesis of RNA
- Forms ribosomes



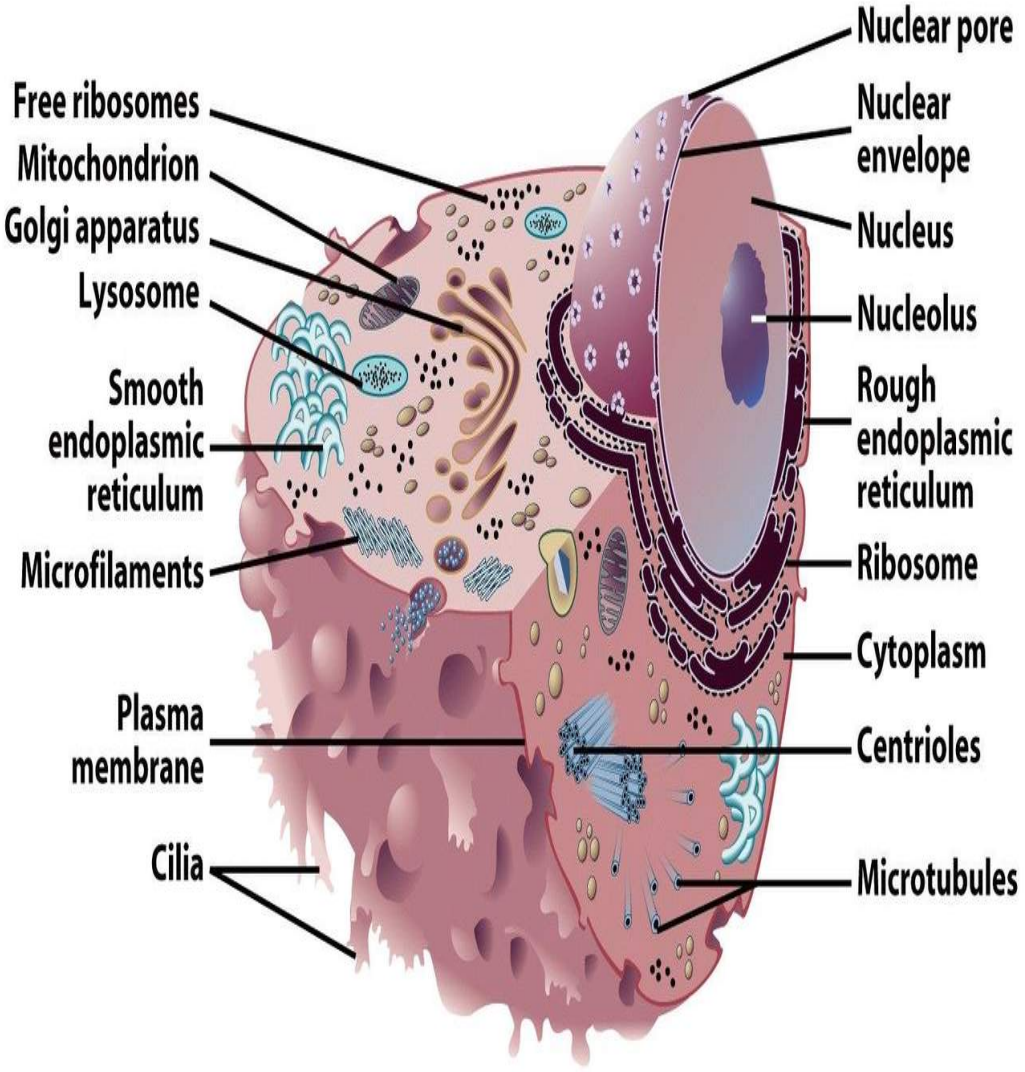
Rough Endoplasmic Reticulum

- Ribosomes attached to surface
  - Manufacture protiens
  - Not all ribosomes attached to rough ER
- May modify proteins from ribosomes



Smooth Endoplasmic Reticulum

- No attached ribosomes
- Has enzymes that help build molecules
  - Carbohydrates
  - Lipids



Golgi Apparatus

- Involved in synthesis of plant cell wall
- Packaging & shipping station of cell

1. Molecules come in vesicles
   2. Vesicles fuse with Golgi membrane
   3. Molecules may be modified by Golgi

4. Molecules pinched-off in separate vesicle

5. Vesicle leaves Golgi apparatus

6. Vesicles may combine with plasma membrane to secrete content

